Mike Levin & Anna Gitelman - Designing with Diverse Intelligence: Interfaces, Agential Materials, and Ingressing Minds

The conversation explores intersections of biology, art, architecture, and design, focusing on regeneration, cognition, and agency. It emphasizes a shift from anthropocentric perspectives to designing environments that collaborate with diverse intelligence: biological, synthetic, and more-than-human. Topics include [illegible] reinterpretation, platonic [illegible]. Examples from bioengineering and architecture illustrate how living systems and built environments can act as adaptive, participatory agents. The dialogue envisions environments of co-creation, merging biological processes and technological systems into dynamic, participatory, and regenerative ecosystems.

Mike Levin & Anna Gitelman were interviewed by Michael Just

MJ I will say just a few words as to why I thought it would be wonderful to have this conversation. My own background is in art, but I have been working with architects and designers since the beginning of my practice. And, Mike, I have been following your work for years now and have started integrating it in my teaching. One thing that I find so exciting about it is to recognize diverse intelligence as a broader field of research. How can we as artists, designers and architects participate in this field, and ask questions that we all share to a certain extent? For me, regeneration has become a paradigm and perhaps it can be said for many fields despite very different implications and [illegible]ction to regeneration. We can [illegible]. I think this also critically touches upon what you call mind blindness. In one of your recent conversations with Pamela Lyon she said that the broader scale of implications of your work is providing a much-needed perspective for us to be better equipped to handle the ecological crisis that we're facing. I would like to use this as a starting point, as a point of connection to see how creative practices can align with your field of inquiry. I'm excited that Mike has made the connection possible and for Anna to join in here. I think your perspective on this is super valuable.

AG I'm coming at this from the creative field of architecture. And I have to say, all those questions that you asked right in your introduction are very relevant and discussed in our field quite a lot. Now, do we all know exactly how to apply the kind of knowledge Mike is exploring? No, definitely not. But that's what makes this conversation so exciting. As designers, people who shape space and build environments, there's a lot we can learn from bioscience and bioengineering. The processes they reveal offer new ways of thinking, and I think there's real potential in exploring how those insights could inform what we do.

ML Pamela is very kind and I just want to be clear that I am making no claims to knowing how to turn my ideas into resolving something as big as the ecological crisis. But I do think that there's something very deep that unites the kinds of things that we're all interested in here, which is, and it's a weird way of putting it, certainly for biologists, but maybe you will recognize it more from your side: I think one important thing that ties all this together is the importance of storytelling, and specifically this idea of doing so continually, which is what I think living things do all the way from the bottom. So, this applies from the molecular networks to the cells, to the tissues that make up your body, all these systems, and of course, the organisms themselves, and probably ecosystems too, I guess. What they're constantly doing is dynamically creating and revising a model of themselves in the outside world. We have this idea about the memories that you're given, so the genetics that your past lineages have passed on to you. There's this kind of standard view where these things are prescriptive of, here's your genome, and this is what you're going to be, and this is what you're going to do, and so on. That's not how it works. And if you want, we can get into some of that. Rather, these things are prompts from your past self, like our memories are, or from past versions of your lineage. And they have to be creatively interpreted at all times to make not the most similar story to what was happening before, but the most adaptive, useful story of what you're going to do next. And this process is, I assume, you guys can tell me, I assume this is something relevant to what artists do in terms of interpreting the things that they've been given in terms of past performance of other people in the natural world around them. To use these things to tell better and more interesting and more adaptive stories into the future. And my point is that all of life does this, from evolution to embryonic development to regeneration. All these kinds of events are basically an ongoing process of sense-making.

MJ There would be a lot to say from my perspective on how we might relate this to art, and perhaps we can get there. For me, art practice is fundamentally regenerative in the way it intends to operate, what it intends to facilitate. And let me expand on what that would imply. But I suppose the question I find so interesting is: regeneration, care and repair, they are ubiquitous at this point in design, art, and architecture discourse, and certainly for good reasons. And yet, I find that the scope of this discourse, the degree of its potential for recognition and integration remains limited. I think this is the implications of diverse intelligence, meaning, how do we step beyond, overcome what Mike, you call mind blindness? How can we design for forms of cognition that go vastly beyond the human? And as you were saying also, diverse embodiments beyond three-dimensional space. How do we integrate diverse embodiments, how do they inform the way that we structure worlds, so to speak? This question, for me, is of paramount importance to design practices as worlding practices. Anna, do you have any thoughts on this, do you see limitations to the extent that I just laid them out? Where would you locate this discourse in terms of your own practice?

AG Well, the idea of mind blindness, I think it starts with the fact that we have a hard time recognizing intelligence if it doesn't look like us. For centuries, design has really worked within that limitation. We've been centering everything around the human as the default subject of spatial and material design. But the truth is, we've always been surrounded by other forms of intelligence: plants, fungi, microorganisms... And their ways of thinking, sensing, and engaging with the world have mostly been left out of how we design and imagine our built environments. But I think that's starting to change. We're beginning to expand our cognitive maps to question that very anthropocentric view of design. And we're seeing some exciting projects that are trying not just to design for other forms of life, but in some cases, to design with them. Take the CookFox office façade, for example, it's a collaboration with Buro Happold. What's interesting about that project is that it integrates microhabitats right into the building's envelope, tiny spaces for birds, insects, and plants. So, it's not just an aesthetic or performative feature; it actually functions as an ecological interface. It provides nesting cavities, supports pollinators, integrates native vegetation, it's environmental stewardship built right into the architecture itself. Another really interesting example is the Hy-Fi Pavilion by The Living, designed for MoMA PS1. This one takes things even further. The whole structure was made from bio-bricks grown from mycelium, the root system of fungi, combined with agricultural waste. These weren't traditional construction materials; they were cultivated in molds. The result was a completely biodegradable structure. It was low-impact and radically non-anthropocentric. And that's a major shift, from thinking about design as a set of instructions imposed on passive matter, to co-creating with biological systems. Both of these projects, to me, point toward what we might call multi-intelligence-centered design. It's a paradigm shift. Architecture starts to not just serve human needs, but to recognize and collaborate with the more-than-human world. And looking ahead, I think - and Michael, you are absolutely right- that the big question becomes: how do we design for forms of cognition that exceed our own understanding? That might mean advanced AI, emergent ecosystems, or even interspecies communication networks. To do that, we might have to give up some control, let buildings evolve on their own, optimize themselves, maybe even co-author their own form through sensors, feedback loops, or responsive materials. Design might become less about authorship and more about stewardship, curating ecosystems rather than controlling them. And maybe buildings stop being just environments for humans and become interfaces for multiple forms of awareness, biological, synthetic, and everything in between. It's a huge shift, but I think it opens up some incredible new ways to think about what architecture can be.

ML Another interesting way to look at this is in terms of this idea of freedom of embodiment. So, one of the things that we're working on is regenerative therapeutics to fix birth defects and reprogram cancer and to regrow missing or damaged structures and things like that. The question is basically: how do I communicate the correct final shape to a set of cells? That's really the bottom line to all of these biomedical needs, is the ability to communicate a final form to groups of cells that they will then build. Once you have that information, you don't have to stick to the original design of current modern humans. You can modify that in many ways. And of course, there is nothing particularly unique or special about today's human anatomy. There are many things that we would want to fix and improve and change. In general, this idea of freedom of embodiment, I think in the future, people will look back on us in the way that we look back on some of our early hominid ancestors and the way they lived, and they would say, wow, it is amazing that these people were just born in whatever random body nature gave them and they had to stay that way their whole life. They had no choice about it. I mean, I think people are going to find that really wild. So, the point is that much like what I said about a month ago to a convention of psychoanalysts, which is that your clientele is about to get a lot weirder and wider than the normal humans, and you're going to have to figure out how to deal with beings that are cyborgs and hybrids and extended and altered in many different ways. I know nothing about architecture or design, but it's very interesting to me to think about how we're going to design those things when the beings for whom we are designing acquire an incredible variety of form and cognition. You can just imagine you've got your standard humans, let's say, and then you're going to have beings that have senses that are about the solar weather and the stock market and the emotional state of their best friend and whatever. These will be primary senses that we have. Like you see in the work of David Eagleman and lots of people in the sensory augmentation and sensory substitution space. And you'll have people with additional limbs and different ways of manipulating their environment. And who knows what kind of chairs they're going to need once their basic body plan is quite different. I'd love to hear you guys talk about that because I don't think it's that far off actually. And I wonder what the world is going to look like when you can't assume that the beings for whom you're building have the same standard structure that we've had for, you know, millions of years.

MJ Embodiment might be pertinent here and, as you point out, Mike, I wonder if we're grasping its full implications. When we think about AI and AI embodiment we seem to revert back to some sort of human-centric notion of things in three-dimensional space, a robot, something that walks around. But we can think of embodiment in ways that go far beyond that, namely I suppose any navigation of option-spaces presupposes a coupling to a specific substrate, a body, so to speak, as you point out. I think this has important implications for design, architecture, cities, everything. Can we find ways of integrating diverse spaces as much as diverse ways of navigating them, and how would this change our very notion of design. Anna, you pointed out an important distinction in terms of “designing with” in contrast to what I said earlier “designing for”. I entirely agree with you, and I think it is indicative of the broader framework of this discussion, the recognition of and collaboration with unconventional forms of and potentials for agency. Where I think that the “designing for” perspective might be nonetheless useful is, to use Mike’s notion of the cognitive light cone, that as humans our potential scope of care and caring for might go beyond that of other forms of cognition and in that lies a specific kind of stewardship and responsibility that is important for creative practices. This might point to a broader framework that I touched upon earlier, of the dynamic between extraction and regeneration. We appropriate, that is, extract from, that which we are blind to, and we can regenerate in collaboration with that which we recognize. And I think that question is very interesting and challenging to navigate. And here, I think, is where Mike’s work comes in and is so critical to this discourse in the creative field.

ML Where I think we're going to have to end up is that, what we see in biology the environment is what it is, and that various beings are really good at adapting themselves, not just on an evolutionary timescale, but on a single individual being's timescale, adapt themselves to the reality of the environment. We have all kinds of examples of plasticity in our lab where you make changes, in fact, sometimes pretty drastic changes, and life finds a way to be a coherent organism that will persist despite all these changes that you're making. I would love to see a future in which the environment, through the efforts of your community, what we are getting at is an environment that actually adapts to the being as well. So, there's agency on both sides. It's not a one-size-fits-all where beings with different cognitive structures have to adapt to an environment that's comfortable for one kind of being, but maybe not at all comfortable for some other kind of being, right? Both physically and mentally. Can you think of a future in which your environment, meaning the space that you live in, the space that you work in, the space that we navigate outside and so on, has elements that actively seek a kind of harmonious collaboration with all kinds of weird beings that are going to be living there? So that it's not just one-directional, as in, now we have to have the plasticity to adjust to whatever the world looks like, but actually we are now building artifacts, both indoors and outdoors, that are meeting you halfway and trying to have a more harmonious interaction.

AG If I can jump in, I'm coming at this from the perspective of space, and more specifically, how we actually inhabit it. Historically, architecture has been shaped by this assumption that all human bodies function in the same way. You know, we walk the same, move the same, perceive space the same. And because of that, design has often revolved around this idea of the "average" human. But I think we're starting to realize that the model is way too limited. As we look ahead, the challenge, and the opportunity, is to create spaces that aren't just human-centered, but adaptable to a much wider range of beings and experiences. And I don't mean just different kinds of people. As Mike mentioned earlier, we might be designing for entities that aren't even entirely human. One area where we're already seeing this kind of rethinking is the experimentation with wearable architecture. These are essentially micro-environments, little spatial systems designed to respond directly to the needs of an individual. They're incredibly adaptive. They don't assume a fixed human type; instead, they respond to the body they serve in real time. Now, what's exciting is that this kind of thinking doesn't have to stop at the scale of the body. It can scale up. Take modular systems, for example, small, self-contained units that can function on their own but also connect, evolve, and serve as testbeds for adaptation. It's a totally different mindset than standardized, one-size-fits-all spaces. So, if we imagine a future shaped by this approach, we might see buildings or even entire city blocks made up of modules, each one unique, each one responding to different users, environments, or needs. But these modules wouldn't be isolated. They'd still be part of a larger, dynamic system, constantly interacting, evolving, reshaping themselves in response to what's around them. That kind of adaptability could completely change the way we think about architecture.

SUrF Speaking of adaptability, the dynamic of compression and expansion, the bowtie architecture, which we find for example in biology but also in machine learning, is very interesting, particularly in terms of this notion of creative interpretation or reinterpretation, and to do so, let's say, together or communally. From an artistic practice perspective, that's where I think art can potentially intervene and find ways or produce a framework, as in the conditions for, let's call it a coming together. So, I think of regeneration in a sense as a kind of creative coming together and then sort of expanding from there. And I think we have to continuously renegotiate that. And that's what really interests me, how to intervene in that regard, and to do so alongside different forms of intelligence. And to see them as creative, as capable of creative expression in their own right. There's this quote from Jung in Mike's "Ingressing Minds" paper: "The artist is ... one who allows art to realize its purposes through him (or her)". We might as well think of artists here as designers or architects, amongst others. That's a controversial quote. I think it's very interesting and I could guess your intentions of including it, but maybe you could say, Mike, what's your take on it in that way?

ML I interpret it in an even more controversial way than one might think, which is that, by artist, I mean cells and tissues and even molecular networks, because all of these things are in a continuous process of navigating various problem spaces. And for reasons we can get into, a lot of what they're doing is not algorithmic, it's creative. And this idea that what they're actually doing is, which I assume artists, scientists and inventors and whatnot are also doing, navigating the space of possible solutions to some problem. The problem might be, how do I survive the next 10 minutes? Or what is the meaning of life going forward and anywhere in between? And this continuous search for solutions in a problem space where there are many things that you could do is something that is faced by particles as they go through what are called least action laws to find the most efficient path to wherever they're going. Then you start building up and the things that we call life, which are these networks that navigate different kinds of problem spaces, all have to do this. They must all search through a space of enormous possibilities. They don't have time to really do an exhaustive search because they will be dead long before that happens. So now, what is the process by which everything from molecular networks for cells up to human artists and various geniuses just sort of find solutions, they're in the flow and they find solutions easily, and the rest of us that are kind of methodically cranking through some process to get there. What are the processes by which all of these beings identify the thing that they're looking for? And one of the directions that we're going into now is actually looking at it from a slightly different perspective to say that the agency is not just on the side of the being who has to sift through an enormous amount of passive data and passive material, but that there are patterns. In that platonic space paper, I argue that some of these patterns are familiar, low agency things like facts of mathematics and so on. But some of them are very complex, high agency things like kinds of minds, and everything in between, that these patterns are actually active in that process. So, it's not that we are ourselves just looking through a bunch of passive patterns, but the patterns themselves have a kind of, we don't really have the vocabulary for it, I don't know if you want to call it a force or a drive or something. But they are actively reaching out. The solution is in some sense reaching out to an appropriate seeker. And again, I don't just mean the human inventor trying to create a majestic work of art. This is for everything. This is all the way down, okay, all the way down through the material of life. And so that's the idea that that there are these, to the biologist, they look kind of like free lunches in the sense that you make something or you engineer something, or evolution evolves a certain embodiment, and it immediately becomes the recipient and the beneficiary of all kinds of amazing facts of mathematics that do not come from the physical world. These are things that are not determined by any facts or truths about the physical world. They're facts of mathematics that come from a different world, but they provide all sorts of amazing advantages for free. Just as a simple example, imagine that in your world, the highest fitness belongs to a particular kind of triangle, okay? So, you're in some world where there's a certain kind of triangle that is the organism with the highest fitness. Evolution will crank through a bunch of generations finding the first angle, and then it'll crank through a bunch more generations finding the second angle, and then it doesn't need to look for the third angle because you know that in flat space, if you have two angles, you know what the third one is. You don't have to find it. Now, where does that fact come from? You didn't have to evolve it. Evolution just saved itself one third of the search time to finish this construction. And it's this magical gift from the laws of geometry that you don't actually have to do any effort to get. You just have to make the conditions right and then here you are. That's the ingression. The ingression is when you build a physical interface you pull down a bunch of these amazing truths of mathematics that then evolution makes use of. So that's what I was going for there. I thought that quote kind of summarized really well this idea that as much as we are searching for solutions to our problems and to various creative works and so on, I think that in an important sense, these things exist separately. And I think they are, in effect, searching for us in a very symmetric way to the way that we are searching for them.

MJ I've read the paper, of course, and I think it's a wonderful, really intriguing notion. Anna, could I ask you, the idea that we locate agency and purpose, so to speak, outside of the traditional author, how would you react to that? Is that something we can work with?

AG You know, I think any creative work really pushes us to ask: where do ideas actually come from? And how do they take shape? What Mike brought up earlier really resonated with me, it challenges that old idea of the creator as this lone source of inspiration, like everything has to come from one mind. I don’t pretend to have all the answers here, but I’ve been thinking a lot about this idea of the physical interface, whether it’s architecture or art, as this point of contact between thought, material, and the environment. And when you look at it that way, creative work isn’t just something that happens in your head. It’s shaped through interactions, with materials, with technologies, with spaces, and maybe even with nonhuman intelligences. And once you start thinking about it like that, the whole idea of authorship begins to shift. It’s less about a single person having a vision and more about working within a network of contributors, where ideas or outcomes emerge through a kind of negotiation across many inputs. To me, that doesn’t take anything away from creativity. If anything, it opens it up. It becomes a much more expansive and generous way of thinking about how we create.

MJ The idea that there is a drive or a sort of affect, for lack of a better term, mediating this physical instantiation … I understand, Mike, you obviously have been talking a lot about Whitehead, and I'm not a philosopher, but I try to engage with these ideas, too, as a way of making sense of this. I understand that, for Whitehead, agency is immanent to the physical world but eternal objects, as he calls them, lure themselves into instantiation. Maybe there's a kind of mediator at play. And yes, the physical interface that Anna just brought up is a very nice concept to think about. I guess we could call it a boundary, a dynamic as much as relational boundary. I would really like to explore further what this dynamic relationality could mean for architecture on a more practical level. I also like the idea that as practitioners, for us the testing ground of these ideas lies in the physical word. What do these perspectives and new possibilities allow us to do differently? The implications of Jung's quote I think are interesting in that they are destabilizing a Western artistic discourse which rests on an author, an artist, and human ingenuity, not to say that these notions haven't been called into question. It's a discourse that seems preoccupied with interiority in that it locates creativity, similar to mind and consciousness, within the individual and thereby fails to account for other, particularly relational ways of their instantiation. If we take Mike's notion of basal cognition, the idea that a trace of cognition can be found everywhere, if I understand you correctly. I would say this is true for art as well and that an argument could be made, let's say from Whitehead via Ashby to Kauffman, for example, that creativity scales according to the potential for or intensity of experience. The most basic adaptive behavior would already be creative in this view. I'm not prepared to take the argument to its full development here, but I suppose that everything might express "art" to a certain extent, meaning the degree of cognition corresponds to the degree of creative expression as potential for transformation. For that, however, we have to give up the binary thinking that something is art, and something is not art, and this is, again, Mike, something I understand you're saying: the differentiation between conscious and not conscious is the wrong way of thinking. In art, in one way or another, we have to make this differentiation, especially in a more contemporary discourse where art is contextually and institutionally based. You bring something into a context, it turns into art, you take it out, it ceases to be art. Of course, we can find exceptions to this and my framing is reductive. It is an important discourse but, for me, it's a lot more interesting and a lot more productive, perhaps also disconcerting in a way, to think that, yes, everything expresses art to a certain extent, regardless of how small it might be, and that's actually a way of thinking, especially a transdisciplinary way of thinking. I would be prepared, just intuitively, to say that for design and architecture as an organizing principle, especially in terms of questions concerning niche construction and self-reinforcing patterns, the same applies, but that's beyond my competency. But Mike's work has really inspired this way of thinking for me: we have to think in gradients and not in this binary paradigm. This seems to be difficult.

ML I'm certainly not going to try to define art or anything like that, but I agree with you on the aspect of creative interpretation. Josh Bongard and I have been developing a system called polycomputing. And it's this idea that I'm sure you guys will recognize, which is that unlike in other areas of computer science and other sciences where you have something and it has one objective meaning and we can sort of argue about what that is, but eventually we get to the right answer and that's it. We all know now what this means. Polycomputing is this idea that you have to focus on the extraction and the creation of meaning by various observers. It's basically that the outcome of any computation and that of any physical process, and the question of what does this thing actually compute, that there is no one correct answer, but rather it's in the eye of the beholder. And I've certainly heard this idea in, let's say, literature and things like that. The author may or may not have had something in mind, but that's not necessarily what it means, where meaning is located. Other readers are free to get something else out of it, that idea. So, from that perspective, every physical system is doing something that can be interpreted by itself and by others around it, other observers. And when I say observers, I don't just mean human scientists, I mean parasites, conspecifics, its own parts, the system of which it is a part. All observers, all up and down scales in biology, all of those are up for grabs and up for interpretation in the most adaptive, useful, creative way that they can. And this is what we see with genomic information. This is what we see with memories, the behavioral memories and other kinds of information, that what living systems or the things that we call living are especially good at is this kind of creative interpretation of what's going on to find a perspective from which this thing is the most interesting and useful about how to live life into the future. And again, you can see it as an extremely practical thing for very simple organisms, and then kind of a very broad, large-scale, lifelong endeavor for humans and beyond. So yes, I agree with you. I think that in a certain sense, what's special about art is happening everywhere, all the way down. And just one other quick thing. You said something very interesting earlier about coming together and how it is that subunits come together. And I just want to point out that that ability to interpret things is critical for coming together in specific ways, and I'm sure you guys have other examples of this. When we look at an early embryo, which might have, let's say, hundreds of thousands of cells, the reason that we say, oh, look, there's one embryo instead of, hey, a hundred thousand cells: What we're actually counting as one, when we look at it, is a story that all of those cells have bought into. They're committed to one story about a path they're going to take through anatomical space. That's what makes it an embryo, is because they are all aligned to the same representation of what the correct pattern is going to be. And they're all going to work together really hard to make that pattern. If you try to intervene and disrupt them, they will find new ways around it. They will do all kinds of clever things to get there. But it's a coherent whole instead of a bunch of parts because of a model of itself that holds them together. So, this idea of stories and models that hold parts together into wholes that have their own separate identity and the ability to project into larger problem spaces and things like that is really critical. This is something that sounds very kind of philosophical and airy and whatever, but we use this in the lab every day as a roadmap to therapeutics. This idea that what you're actually manipulating when you want to regrow a limb or normalize a tumor or fix a birth defect, what you should be actually manipulating is not the genome, not the molecular pathways, not the cells, which is what modern molecular medicine is focused on, but what we can actually manipulate are the stories and ways in which these cells actually buy into these things. And if you want, I can give you some very specific examples about how we do this. But that's really the driving principle. These cognitive glue mechanisms that bind subunits towards a particular outcome, that's an amazingly important and tractable target if you want the complex system to do something different than it was doing before.

AG Yeah, it's super interesting. I think the idea of polycomputing is really compelling. From where I'm coming from, architecture, I actually see the built environment as kind of a polycomputational interface. It's not fixed; it's something that invites multiple interpretations depending on who's engaging with it. You know, a space never just has one function or one meaning. That meaning shifts based on the observer, the context, the moment in time. I would also like to comment on the idea that living systems can be both interpreters and interpreted. Architecture's the same. It's never neutral. It acts, and it's acted upon. Think about the basic function of architecture of being a shelter. Depending on who's looking at it, even this basic function can change, it could still be a shelter, a place of safety and rest for some species, or a thermal regulator, or even a source of hydration for others. So, the same structure means very different things depending on who or what is encountering it. That makes me think of Derrida and his concept of *différance*, where meaning is never fixed but always shifting, always deferred, depending on context and the whole chain of signs and references around it. If we apply that to architecture, suddenly a space isn't just a static container anymore. It becomes something performative and generative. Its meaning is shaped by time, user, species, and use. Just like Derrida said about language, it never really settles into one meaning, I think architecture doesn't either. It's always evolving, always being reinterpreted. It's experienced differently, valued differently, depending on who, or what, is engaging with it. And that, to me, opens up so many possibilities for how we think about space and design.

MJ That's a very interesting point regarding the shelter. The dynamic between shelter, interface and boundary, it touches upon Mike's work on the computational boundary of a self and, I think, more broadly speaking, Free Energy Principle accounts of separating internal states from external states. I really like this quote from Richard Watson where he differentiates between the static notion of "what exists, persists", and the generative potential of "what relates, creates". This actually brings us back to Derrida, in the sense that différance shows why the metaphysics of presence is a fallacy because what persists is continually reproduced by relational differences. And I think what you were just saying, Mike, in a sense, is that there is a persistence based on a shared story but that this is dependent on this continual creative aspect of relating. I suppose the conditions of the shared story, where would you locate that? Anna, I was just thinking of architecture as a shared story, actually.

ML First of all, on the persistence aspect, there are always multiple observers and the system itself persists for itself in specific ways, but different observers may or may not see it. So just as an example, if you have a soliton wave or a hurricane or anything like that, if an observer is able to pull back to an appropriate scale of space and time, they can see a system persisting for some amount of time. If they zoom in and they're hyper-focused on the molecules of air or water, they're not going to see any kind of large-scale persistent object because the matter keeps moving in and out. When you have a wave moving through an excitable medium, it's not the material that moves. If you're focused on a particular piece of water, let's say you see it moving up and down and that's the end of that. What you don't see is this persistent wave that actually travels across. up to the observer, actually. And this goes back to our point about truly diverse intelligence. It's always an IQ test for the observer to say, do you see what's actually persisting here and what level of intelligence it has? Because we may be focused on entirely the wrong thing. And just be completely blind to it. I tell this fictional story sometimes to illustrate this idea. Imagine these beings that live in the core of the earth. And they're incredibly dense, and they've got gamma rays for vision and whatnot. They come up to the surface, what do they see? Well, they don't see us, and they don't see any of the physical objects, the tangible stuff around us. We are a thin gas to them, a thin plasma. They're so dense that all of us are basically in this barely detectable phase. But if they have the tools, the equipment to detect it, they will see patterns in that gas that hold together for about a hundred years. And they sort of look like they're kind of doing things, but you can easily forgive them for thinking that, well, patterns in a gas can't be agents: we are real physical agents, real beings. These things around the Earth's surface are just patterns in the gas. They're cool and all, but they're like waves in the medium. You can't really take them very seriously as independent agents. And so it's really up to the observer to have both the physical capabilities and then the mental structure to actually be able to observe these other beings. And just as a specific example of how these stories are implemented, so think about, in this case, a frog embryo, but it's basically the same in all others. The question of why are your eyes located where they are? Well, earlier, as the head starts to be formed, there's a very particular bioelectrical pattern within the tissue that says, build an eye here. And what we've learned to do is A, detect that pattern, and B, to recreate that pattern somewhere else. So, what we can do is go in and create: and this is not by applying electric fields or anything like that. We have very much more subtle ways of manipulating the electrical properties of cells, the way that they manipulate each other's electrical properties. And we can set that, we can create that voltage pattern somewhere else, let's say in the tail of the embryo and say make an eye here. So, what happens then is very interesting. We can manipulate a few cells, and what those cells do, they realize immediately that there's not enough of them to build a complete eye. And what they do is they tell their neighbors, hey, you got to help us build an eye. They try to recruit their neighbors. And we can see this because we know exactly which cells we've manipulated. And we know exactly which cells we did not touch. And they also participate in the formation of an ectopic eye. And you get an eye on the tail of a tadpole. But what's actually happening is that those cells are telling their neighbors, we are all going to make an eye. So, we force them into this story that there should be an eye here, they've bought in, they are going to then convince their neighbors, we don't need to worry about that as engineers, because we're dealing with this amazing agential material that has the ability to do that. We didn't have to teach them to do that, they already do that. So, we insert this new idea that make an eye here, that's a particular bioelectrical pattern that basically specifies the meaning of what we're trying to say. They try to convert their neighbors. But their neighbors, as part of an ancient cancer suppression mechanism, aren't just sitting there, they resist. So, the neighbors are saying the opposite, they're saying to these cells, no, you should be skin. And so there's a battle, what's happening is a competition between two worldviews, so to speak. And I'm purposely using these kinds of words because we're talking about these larger human issues. What you're looking at here is the battle of two models of what our future should look like. You've got a set of agents that say, it should look like this, it should be an eye. You've got a bunch of other agents that say, no, you're an aberrant growth and we're going to try to equalize you out and make you like us, which is skin. And depending on how you do it, sometimes the one wins and sometimes the other wins. Sometimes you see that the eye just barely starts to get going and then the other cells, they normalize it, they wipe it out and you get nothing. And that's when the skin model has won. And other times the eye idea takes hold and it sort of captures a region and boom, you've got this eye on the tail and will eventually end up with a frog with an eye on its butt. And these eyes are functional, by the way. So, we have the ability to read, and then to inject modified versions of these models of themselves and of the future into tissue.

Our control is still very poor, but ultimately, when we get really good at this, this is the roadmap to regenerative medicine. You can tell cells, build whatever, to the extent that we are convincing, meaning that we are able to induce the right amount of plasticity. And so we're working with plasticines and other things to sort of open up the possibility space, because what kind of messages you're willing to hear is very much part of your priors and your history. Some of these cells, the things that give rise to scarring and other things that are non-regenerative in mammals, they're not particularly natively open to these other signals, but you can make them more so. So that's kind of a roadmap to creating the construction that you want, is by changing the story, not the hardware, the story that they're going to follow.

MJ This perspective that you just outlined of agential patterns is really transformative. Elsewhere you quoted William James in that "thoughts are thinkers", which might be worth bringing up here for context. It shows that what is an agent and what is possibly agential, that agency and the potential for transformation, actually, as we said the potential for creative expression, can manifest in very unconventional ways that are not obvious to us at all. This also touches upon what Anna said earlier about architectural structures and multiple functions, the openness to interpretation, perhaps the openness to different ways of activation. From an art and design perspective, how do we think about diversity and sameness, the common, as we might refer to it, what someone like Joscha Bach talks about as shared purpose? Maybe shared purpose is something that, Mike, you have touched upon in what you just said, and your example of cells is really wonderful and fascinating. As I said, for me, the question of negotiating this dynamic of diversity and commonality in light of shared purpose is really profound in terms of what artistic or design practices could, let's say, aspire to create the conditions for. I see this in a way as a potential reconciliation of purpose and the sacred, in Batesonian terms, the sacred as interconnectedness. And I think there again, I see some overlap, at least conceptually, with the emphasis on a common set point that is flexible and negotiable to an extent, rather than a fixed genomic library or origin story, as in an essence or an essentialist approach. But, Mike, you already outlined this earlier, the determinism of "this is what you are, and this is what you're going to do" doesn't hold.

ML The big thing here is that in biology, and I assume soon, if not already, in architecture and design, what you're dealing with in an agential material requires a completely different kind of engineering. So, for thousands of years, we've engineered with passive matter. And the upside is that it doesn't provide too many surprises. The downside is that basically all it knows how to do is hold its shape. And if you want to make it do something, you're in charge of every aspect of it. If you want to build a simple machine, you're in charge of everything. Once you start working with, then we had active matter and computational matter. And then of course, in biology, we have these agential materials where the material itself has an agenda. It has learning capacity. You can't engineer with it by micromanaging it the way you would with a passive material, you're much more in the land of the behavioral sciences where what you're actually doing is reward and punishment, convincing collaboration with the material. And I'd love to hear you guys talk about what that might look like on the architecture and design side. Another perspective by the way is that we barely have any drugs that actually fix whatever they're designed to, you know, besides antibiotics and a couple of other things. They mainly just try to force the molecular state. They don't actually repair, and oftentimes you get side effects and limited efficacy because the material fights back. And we're only scratching the surface now of learning how to reset the set points and basically get the buy-in of your material to shift to a different state. I think that that requires a completely different way of building.

AG Yeah, that's a really fascinating idea. I think in architecture and design we've traditionally taken a very top-down approach. You know, we come in with a clear idea of how something should look, how it should function, and then we set rules and follow them. But the way Mike was describing it really flips that thinking. It makes you realize that not everything in a system necessarily follows the same rules, or even needs to. I'm particularly interested in the idea that some components of a system may hold their own kind of intelligence. If we begin to see the built environment not as an assembly of passive matter, but as a dynamic constellation of elements, each with its own degree of autonomy and agency, we open up new possibilities. These elements might operate independently, but still align toward a shared goal. That's where I see the analogy with cellular systems. Cities, in a way, already work like this, individual parts doing their own thing, but still participating in a larger system. And maybe if we started designing with that mindset, treating architecture not just as passive matter we control, but as something with potential agency, we'd open up a whole new way of thinking.

MJ I take these points about shared purpose, adaptation, and intelligence as a reminder that a really exciting as much as challenging implication of this conversation is to move from ideas to their implementation in the physical world, beyond metaphor and surface, as we see it so often. How do we operationalize this? And I also understand that we're reaching the end here. I'm super appreciative of your time and of you joining in here. The last thing I would say, and this goes way beyond our limit here, is concerning the notion of ingression. Again, I'm far from an expert but I suppose that incompleteness plays a role here, in that eternal objects are not essences but potentials. Could we think about eternal objects, ingression and locality as in the idea that this manifests differently according to, let's say, certain physical, cultural, historical conditions. The dynamic of ingression, could that be dependent upon the condition of its instantiation?

ML I would refer, again, to this idea of observer perspective. I think the ingression does what it does, but it's up to us as observers to recognize it and to notice it and to use it as a creative prompt or an affordance for our activities. Thank you so much for having this discussion and for letting me think about these wild things that are completely outside my standard remit. Thank you, Mike. And thank you, Anna, so much.

AUTHOR BIOGRAPHY

Michael Levin is the Vannevar Bush Distinguished Professor in the Biology department at Tufts University and serves as director of the Allen Discovery Center at Tufts and the Tufts Center for Regenerative and Developmental Biology. He is also associate faculty of the Wyss Institute at Harvard Medical School and co-director of the Institute for Computationally Designed Organisms with Josh Bongard. His background is in computer science and biology, and his group works at the intersection of developmental biophysics, computer science, and cognitive science. He is primarily interested in how intelligence self-organizes in a diverse range of natural, engineered, and hybrid embodiments. Levin has been developing a framework for recognizing and communicating with unconventional cognitive systems; applied to the collective intelligence of cell groups undergoing morphogenesis, these ideas have allowed the Levin lab to develop new applications in birth defects, organ regeneration, and cancer suppression. His lab also produces synthetic life forms, such as Xenobots and Anthrobots, as exploration platforms for patterns of form and behavior in space of possibilities that in-forms systems from simple algorithms to complex animal life.

Anna Gitelman is an Associate Professor in the Department of Art and Design at Suffolk University. Her research explores the human experience in built environments, with a focus on adaptive environments, modular systems, and emerging forms of social collectivity and architecturally driven communities. She is an active member of professional organizations and frequently speaks at architectural and interior design forums, contributing to the evolving discourse around spatial adaptability. Anna holds a graduate degree from Harvard University's Graduate School of Design and an undergraduate degree in Architecture and Urban Planning from the Moscow Architectural Institute. She has also taught and served as a design critic at the University of Tennessee, CEU San Pablo in Madrid, MassArt, Wentworth Institute of Technology, and Harvard GSD.

Michael Just is a transdisciplinary artist, founder of Michael Just Office & Studio, Berlin, and PhD candidate at the City University of Hong Kong, School of Creative Media. In 2024, he was a guest researcher at TU Delft, Faculty of Architecture and the Built Environment, as CityUHK Research Activities Fund Fellow. He is also a Steering Team member at DigitalFUTURES where he co-organizes the Doctoral Consortium on Architecture & Philosophy and Artificial Intelligence. Michael holds MFA degrees from both the Kunstakademie Düsseldorf (Prof. Daniel Buren) and Goldsmiths, University of London, and participated in the Whitney Museum of American Art Independent Study Program in New York City. Recent teaching includes institutions such as Hong Kong Baptist University AVA and the China Academy of Art, Hangzhou.